EXPERIMENT-9

25/9/24. SUBNETTING

AIM:

Implementation of subnetting in Cisco Packet Tracer simulation.

INTRODUCTION:

Subnetting is the process of dividing a large IP network into smaller logical networks called subnets.

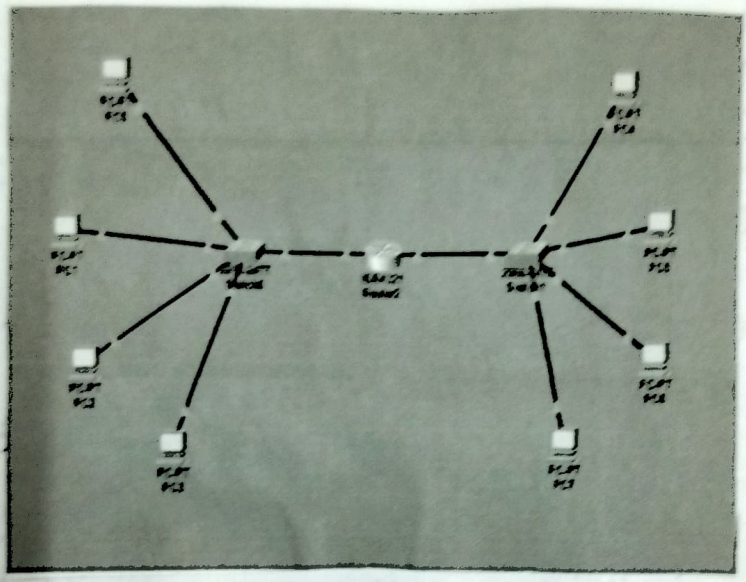
PROCEDURE:

1) Open Cisco Packet Tracer.
2) Add required routers, switches, and PCs
3) Connect the devices using appropriate cables.
4) Configure the routers with the required IP addresses and subnet masks.
5) Configure the router interfaces using the CLI:-
   - enable
   - configure terminal
   - select the required FastEthernet/GigabitEthernet Interface
   - Assign the IP Address and subnet mask
   - Use no shutdown to activate the interface.
6) Configure the switch ports in access mode where required.
7) Assign static IP Addresses, subnet masks and default gateways to the PCs.
8) Use the /27 subnet mask: 255.255.255.224.
9) Configure the four subnets:
   - 192.168.0 | 27
   - 192.168.2 | 27
   - 192.168.3 | 27
   - 192.168.4 | 27

10). verify the router interfaces and PCs between to their respective routers.

11). open command prompt in the PCs

12) Use the ping command to test connectivity between PC and respective router.



RESULT:

Thus, Subnetting was successfully implemented using cisco packet tracer.